

HIGH ACRYLONITRILE NBR: PATENT LANDSCAPE ON CONTROLLED POLYMERIZATION, COMPOSITION CONTROL, AND BATTERY-RELATED USES

1. ABSTRACT

This report synthesizes a patent set relevant to high-acrylonitrile nitrile butadiene rubber (NBR) and closely related nitrile-containing elastomeric or copolymeric systems. The evidence base is broader than classic NBR synthesis alone: several patents address controlled emulsion or free-radical polymerization methods that can be applied to acrylonitrile-containing block or random copolymers, while others focus on downstream formulations where nitrile-rich polymers are used as binders, dispersants, or impact modifiers. Across the set, the target material is defined by elevated nitrile content and by the need to control monomer ratio, molecular weight, and processability without drifting into excluded hydrogenated or carboxylated variants unless explicitly claimed.

The strongest synthesis-oriented disclosure is US7592409B2, which teaches emulsion polymerization with in-situ emulsifier formation and controlled radical polymerization agents, including RAFT, NMP, ATRP, and degenerative transfer approaches. It explicitly contemplates blocks containing acrylonitrile or methacrylonitrile with conjugated diolefins such as isoprene or butadiene. US20100125122A1 adds a multistage mediated-free-radical-to-free-radical process, with optional chain transfer agents and initiators, and specifically identifies styrene/acrylonitrile/rubber systems as preferred embodiments. US9650463B2 shows RAFT-based block copolymerization involving acrylonitrile monomer units, but in a triglyceride-containing thermoplastic block copolymer context rather than conventional NBR.

The formulation-oriented patents are equally informative. EP3605675B1 defines a hydrogenated nitrile-containing binder composition for non-aqueous secondary battery positive electrodes, with nitrile content of 20-60 mass% and a total conjugated diene plus alkylene content of 40-80 mass%, and emphasizes metal-ion-assisted coagulation/hydrogenation and solution turbidity control. US11827830B2 discloses a nitrile-containing copolymer dispersant with 15-50 mass% nitrile units and 40-85 mass% aliphatic hydrocarbon units, targeting conductive material dispersion. US10146149B2 explicitly states that acrylonitrile-butadiene rubber is preferred for charging-member elastic layers. US9306208B2 and US11352449B2 are application-driven but still

reinforce the utility of nitrile-rich or NBR-derived polymers in electrochemical and impact-modification contexts. Overall, the landscape favors controlled radical or emulsion routes, careful monomer-ratio selection, and post-polymerization handling designed to preserve dispersion, adhesion, and molecular-weight balance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 2

- Patent Number: US20100125122A1
- Patent Title: Process for the production of a (co)polymer composition by mediated free ...
- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 3

- Patent Number: US11827830B2
- Patent Title: Dispersant, conductive material dispersion and slurry for electrode film
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 4

- Patent Number: US11352449B2
- Patent Title: Graft copolymer, and thermoplastic resin composition using same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 5

- Patent Number: EP3605675B1
- Patent Title: Binder composition for non-aqueous secondary battery positive electrode, ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2022
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 6

- Patent Number: US8829122B2
- Patent Title: Polyurethane polymer based on an amphiphilic block copolymer and its use as ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 7

- Patent Number: EP2057205B1
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 8

- Patent Number: US10146149B2
- Patent Title: Charging member, electrophotographic process cartridge, and electrophotographic ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 9

- Patent Number: US9306208B2
- Patent Title: Manufacturing method of an electrode for electrochemical element and an ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 10

- Patent Number: US9650463B2
- Patent Title: Thermoplastic elastomers via reversible addition-fragmentation chain transfer ...
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7592409B2 and US20100125122A1 are the most synthesis-centric disclosures: both rely on radical-based polymerization control, but US7592409B2 emphasizes emulsion polymerization with in-situ emulsifier formation and controlled radical agents, whereas US20100125122A1 uses a multistage mediated-free-radical stage followed by conventional free-radical conversion.
- US7592409B2 explicitly includes acrylonitrile or methacrylonitrile with isoprene or butadiene in block architectures; US20100125122A1 explicitly prefers styrene/acrylonitrile/rubber systems; US9650463B2 includes acrylonitrile monomer among many monomer A options but in RAFT-derived block copolymers with triglyceride monomer B rather than a conventional NBR backbone.
- US7592409B2 lists multiple control chemistries (RAFT, NMP, ATRP, degenerative transfer) and in-situ emulsifier generation; US20100125122A1 lists stable free-radical precursors and optional chain transfer agent, decomposition catalyst, and initiator; US9650463B2 specifically requires a free-radical initiator and chain transfer agent in RAFT polymerization.
- EP3605675B1 and US11827830B2 are composition-focused rather than synthesis-focused: EP3605675B1 specifies a hydrogenated polymer with 20-60 mass% nitrile units and 40-80 mass% combined diene/alkylene units, while US11827830B2 specifies 15-50 mass% nitrile units and 40-85 mass% aliphatic hydrocarbon units.
- EP3605675B1 is the only patent in the set that gives a metal-ion concentration window, 50-3500 mass ppm, and links it to coagulation and hydrogenation behavior; US11827830B2 instead emphasizes weight-average molecular weight of 5,000-400,000 and dispersion performance.

- US10146149B2 is the clearest downstream-use disclosure for conventional NBR, stating that acrylonitrile-butadiene rubber is preferred for the electroconductive elastic layer of a charging member; it does not provide synthesis parameters but confirms industrial relevance of NBR in elastomeric formulations.
- US9306208B2 and US11352449B2 are not direct NBR synthesis patents, but they show adjacent application spaces where nitrile-containing or rubber-modified polymers are used in electrochemical electrodes and graft copolymer impact-modification systems, respectively.
- Historically, the older disclosures in the set lean toward emulsion or mediated radical control for architecture control, while the newer disclosures emphasize property windows for battery dispersions and binders, especially nitrile content, turbidity, and molecular-weight ranges.
- Across the set, recurring themes are nitrile-rich composition control, radical-polymerization process control, and downstream performance tuning; the main differences are whether the patent targets polymer synthesis (US7592409B2, US20100125122A1, US9650463B2) or formulation/property optimization (EP3605675B1, US11827830B2, US10146149B2).

4. CONCLUSION

The evidence supports high-acrylonitrile NBR development through controlled radical or emulsion polymerization, with the most relevant synthesis guidance coming from in-situ emulsifier emulsion control and multistage mediated/free-radical processing. For a conventional high-ACN NBR target, the most defensible design space from this patent set is a nitrile-rich copolymer with tightly managed monomer feed ratio and molecular weight, followed by coagulation or formulation steps that preserve dispersion and processability. The strongest composition windows in the evidence are nitrile content in the 20-60 mass% range for battery binders and 15-50 mass% for nitrile-containing dispersants, with combined diene/alkylene content balancing flexibility and polarity. For synthesis, the patents favor radical control agents, optional chain-transfer agents, and emulsion-based processing rather than uncontrolled bulk routes. Practically, the best-supported approach is controlled emulsion polymerization with acrylonitrile-rich feed, a conjugated diene comonomer such as butadiene or isoprene, and molecular-weight control via RAFT or related controlled-radical methods, followed by application-specific post-treatment. The set also indicates that high-ACN NBR is especially valuable where polarity, dispersion, and adhesion are critical, including battery binders, conductive dispersants, and elastomeric charging-member layers.

5. REFERENCES

- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
- US20100125122A1 | Process for the production of a (co)polymer composition by mediated free ... | Not disclosed | US | 2010 | <https://patents.google.com/patent/US20100125122A1/en>
- US11827830B2 | Dispersant, conductive material dispersion and slurry for electrode film | Not disclosed | US | 2023 | <https://patents.google.com/patent/US11827830B2/en>

- US11352449B2 | Graft copolymer, and thermoplastic resin composition using same | Not disclosed | US | 2022 | <https://patents.google.com/patent/US11352449B2/en>
- EP3605675B1 | Binder composition for non-aqueous secondary battery positive electrode, ... | Not disclosed | EP | 2022 | <https://patents.google.com/patent/EP3605675B1/en>
- US8829122B2 | Polyurethane polymer based on an amphiphilic block copolymer and its use as ... | Not disclosed | US | 2014 | <https://patents.google.com/patent/US8829122B2/en>
- EP2057205B1 | Production of meta-block copolymers by polymer segment interchange | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2057205B1/en>
- US10146149B2 | Charging member, electrophotographic process cartridge, and electrophotographic ... | Not disclosed | US | 2018 | <https://patents.google.com/patent/US10146149B2/en>
- US9306208B2 | Manufacturing method of an electrode for electrochemical element and an ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9306208B2/en>
- US9650463B2 | Thermoplastic elastomers via reversible addition-fragmentation chain transfer ... | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9650463B2/en>